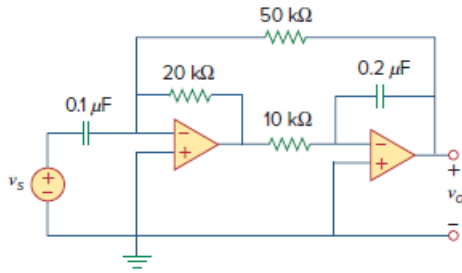


Problem

Obtain $v_o(t)$ for the op amp circuit in Fig. 10.123 if $v_s = 12 \cos(1000t - 60^\circ)$ V.

Figure 10.123:



Step-by-step solution

Step 1 of 7

Refer to Figure 10.123 in the textbook.

The supply voltage is $v_s = 12 \cos(1000t - 60^\circ)$ V.

Compare the equation of supply voltage with general equation of voltage, that is,

$$v = V_m \cos(\omega t + \phi).$$

$$V_m = 12 \text{ V}$$

$$\omega = 1000 \text{ rad/s}$$

$$\phi = -60^\circ$$

Express the supply voltage in phasor form.

$$\mathbf{V}_s = V_{sm} \angle \phi$$

Substitute 12 V for V_{sm} and -60° for ϕ .

$$\begin{aligned} \mathbf{V}_s &= 12 \angle (-60^\circ) \\ &= 12 \angle -60^\circ \text{ V} \end{aligned}$$

Step 2 of 7

Determine the equivalent impedance of the $0.1 \mu\text{F}$ capacitor.

$$Z_{C1} = \frac{1}{j\omega C_1}$$

Substitute 1000 rad/s for ω and $0.1 \mu\text{F}$ for C_1 .

$$\begin{aligned} Z_{C1} &= \frac{1}{j(1000)(0.1 \times 10^{-6})} \\ &= -j10 \text{ k}\Omega \end{aligned}$$

Step 3 of 7

Determine the equivalent impedance of the $0.2 \mu\text{F}$ capacitor.

$$Z_{C_2} = \frac{1}{j\omega C_2}$$

Substitute 1000 rad/s for ω and $0.2 \mu\text{F}$ for C_2 .

$$\begin{aligned} Z_{C_2} &= \frac{1}{j(1000)(0.2 \times 10^{-6})} \\ &= -j5 \text{ k}\Omega \end{aligned}$$

Step 4 of 7

Redraw the circuit as shown in Figure 1.

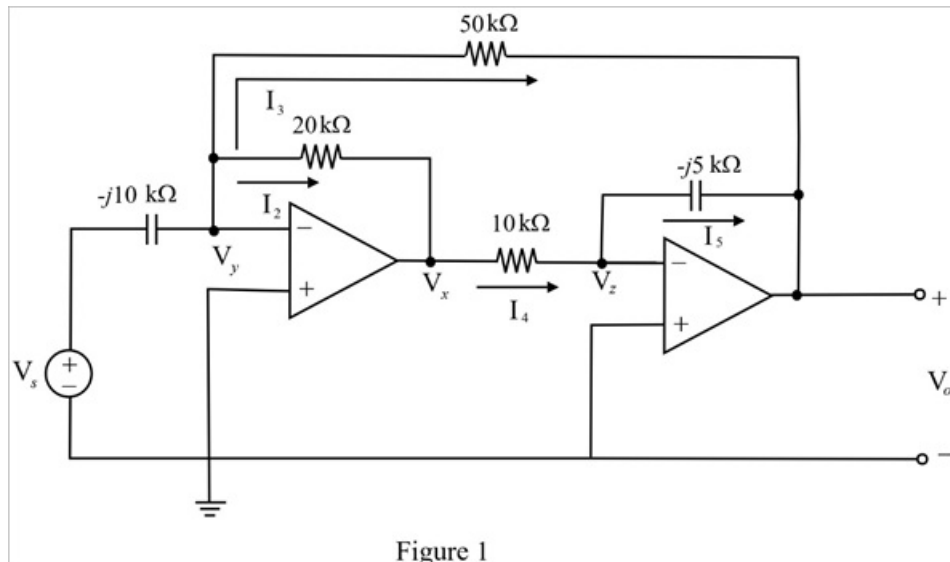


Figure 1

Step 5 of 7

Apply Kirchhoff's current law at node V_y .

$$\begin{aligned} I_1 &= I_2 + I_3 \\ \frac{V_s - V_y}{-j10 \times 10^3} &= \frac{V_y - V_x}{20 \times 10^3} + \frac{V_y - V_o}{50 \times 10^3} \end{aligned}$$

Substitute $V_y = 0$ and $V_s = 12 \angle -60^\circ$.

$$\begin{aligned} \frac{12 \angle -60^\circ}{-j10^4} &= \frac{-V_x}{20 \times 10^3} + \frac{0 - V_o}{50 \times 10^3} \\ 12 \angle 30^\circ &= -\frac{V_x}{2} - \frac{V_o}{5} \end{aligned}$$

$$5V_x + 2V_o = -120 \angle 30^\circ \dots\dots (1)$$

Step 6 of 7

Apply Kirchhoff's current law at node V_z .

$$I_4 = I_5$$

$$\frac{V_x - V_z}{10^4} = \frac{V_z - V_o}{-j5 \times 10^3}$$

Substitute $V_z = 0$ and $V_x = -91.9 - j80.78 - 2V_o$.

$$\frac{V_x - 0}{10^4} = \frac{0 - V_o}{-j5 \times 10^3}$$

$$V_x = -j2V_o$$

Step 7 of 7

Substitute $-j2V_o$ for V_x in (1).

$$5(-j2V_o) + 2V_o = -120 \angle 30^\circ$$

$$V_o = -\frac{120 \angle 30^\circ}{2 - j10}$$

$$V_o = -\frac{120 \angle 30^\circ}{2 - j10}$$

$$V_o = 11.767 \angle -71.31^\circ \text{ V}$$

Therefore, the output voltage, V_o is $11.767 \angle -71.31^\circ \text{ V}$.